Satellite–Ground Collaborative Learning: An Empirical Study on Remote Sensing Vision-Language Model

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***Abstract*—Severe on-orbit computing and power constraints necessitate aggressive compression of ground foundation models for satellite deployment, causing degradation of representation capability. This leaves compressed models vulnerable to out-of-distribution (OOD) scenarios arising from factors like dynamic Earth systems, the diversity of subcategories, and unpredictable user queries. However, the current learn-then-deploy paradigm lacks mechanisms for autonomous OOD adaptation. To address this, we propose the satellite–ground collaborative learning (SGCL) framework, enabling asynchronous, lifelong updates between ground and satellite models. SGCL achieves this through three core mechanisms: 1) on-orbit OOD detector for bandwidth-efficient data downlink; 2) ground-based LoRA fine-tuning triggered by satellite-to-foundation output divergence; and 3) adaptive distillation that aggregates multisatellite knowledge. For comprehensive evaluation, we propose the SGCL-Benchmark that simulates cyclical workflows across seven satellites with specialized protocols for domain/fine-grained/format OOD robustness. Experiments demonstrate SGCL's effectiveness in mitigating compression-induced degradation. Our codes and dataset will be released after acceptance to promote the development of satellite intelligent applications.**

***Index Terms*—Benchmark, out-of-distribution (OOD), remote sensing, satellite–ground collaborative, vision-language model (VLM).**

I. Introduction

EQUIPPING satellites with intelligent models enables real-time analysis, processing, and autonomous decision-making on remote sensing data, significantly boosting application efficiency in areas such as disaster monitoring [1], ocean monitoring [2], environment monitoring [3], and business application [4].

However, as remote sensing models trend toward larger scales and greater functional diversity [5], their computational demands increase exponentially. Meanwhile, due to the limitations of physical size and power supply, the satellite platform faces a computing power bottleneck [6]. This contradiction forces the satellite-based model to be a highly compressed version of the ground-based original. The model compression process significantly impairs its robustness to out-of-distribution (OOD) scenarios [7]. As illustrated in Fig. 1, the compressed model, constrained by limited representational capacity, critically relies on distributional consistency between the training data and the on-orbit environment. In actual space missions, input data frequently deviate severely from the training distribution due to dynamic scene evolution or abrupt user command changes, leading to a sharp deterioration in prediction reliability [6].

Currently, all satellite-based models follow a learn-then-deploy paradigm; parameters are fixed after ground-based training before launch. Updates are solely dependent on occasional manual intervention. This inherently prevents the system from adapting to time-evolving data drift. We argue that the core limitation of on-orbit remote sensing models stems from *the fundamental mismatch between the dynamic spatial–temporal environment and static, closed-loop training strategies*.

We categorize the OOD scenarios into specific categories such as spatial shift, modality heterogeneity, temporal drift OOD, domain, fine-grained, and format OOD. To mitigate the effects of spatial shift, modality heterogeneity, and temporal drift OOD on compressed models, existing studies construct large-scale datasets spanning diverse geographical regions, sensor modalities, and temporal variations. This manner has demonstrated significant robustness improvements against these distribution shifts. However, there are still some profound OOD challenges, particularly domain and fine-grained OOD inherent to sensory data, combined with format OOD risks when user interaction interfaces are present, exposing a critical vulnerability in on-orbit models next.

1) *Domain OOD*: The unbounded complexity and dynamism of the Earth system (e.g., disaster-induced instantaneous surface mutations and unexplored extreme

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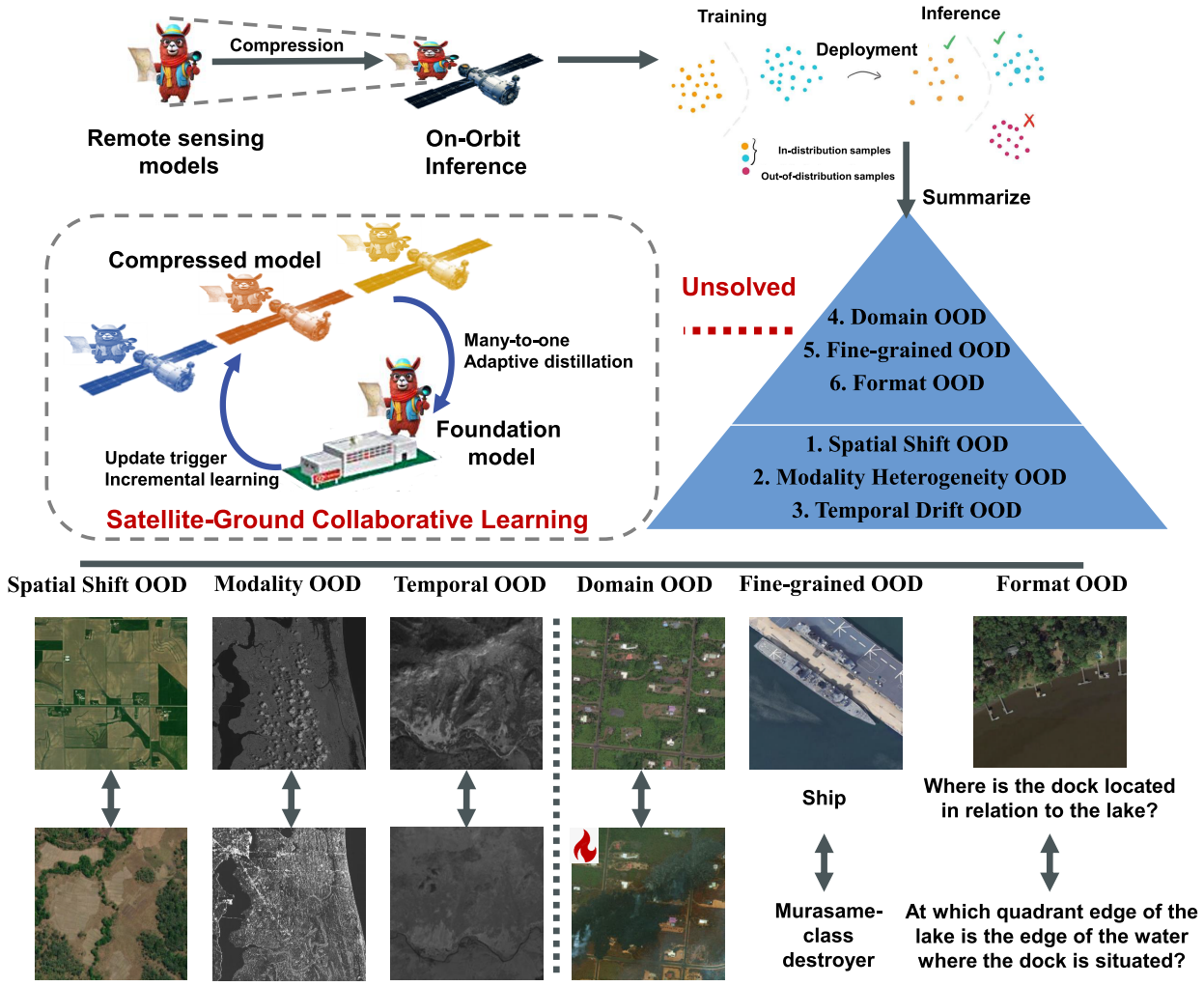


Fig. 1. Main idea of our work.

environments) prevent training data from ever fully spanning the entire real-world semantic space, resulting in cognitive discontinuities in open scenarios.

2) *Fine-Grained OOD*: The inherent internal diversity of remote sensing target categories (e.g., crop varieties, building subtypes, and vegetation communities), arising from growth stages, environmental stress, or new variants, cannot be comprehensively captured by limited annotation granularity and sample sizes. This causes models to fail in generalizing to unrepresented subclasses.
3) *Format OOD*: Satellite-based models that accept user input (like interpreting complex prompts) can fail when given prompts with unusual formats. For instance, nonstandard phrasing, ambiguous task specifications, or syntax not encountered during training. These deviations render static models incapable of interpreting unforeseen instruction paradigms, leading to task execution failures beyond their predefined scope.

The defining characteristics of these OOD challenges are their sporadic occurrence and unpredictability, i.e., a specific task might remain dormant for extended periods or activate abruptly. During model compression, knowledge corresponding to these low-frequency scenarios is often prioritized for removal due to its minimal contribution to short-term accuracy [7]. Conversely, the original ground-based model acts as a knowledge repository, preserving precisely these critical long-tail representation capabilities. Current satellite-based models, lacking both this precompression knowledge reservoir and the capability for autonomous on-orbit updating, find themselves dually disadvantaged when confronting profound OOD scenarios.

These challenges collectively underscore the need for a new paradigm. Satellite-based models must possess the capability for on-orbit autonomous evolution, enabling them to dynamically optimize their parameters based on the task performance and thereby resolve the mismatch between static training and the dynamic environment. To this end, we propose the satellite–ground collaborative learning (SGCL) framework that enhances the satellite's on-orbit model's ability to handle OOD challenges through asynchronous updates between the ground-based foundational model and the on-orbit model. This mechanism also enables the ground-based foundation model to aggregate personalized knowledge from multiple on-orbit models, thereby achieving lifelong learning capabilities.

Concretely, the on-orbit model is a compressed version of the foundational model. To overcome severe satellite–ground

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bandwidth limits, we propose an on-orbit OOD monitoring module as a transmission filter. This module only triggers downlinks of batch data and model predictions when detecting significant distribution shifts or semantic drift. Upon receiving an OOD alert and corresponding data, the ground station's update trigger module quantifies the weighted KL divergence between satellite and foundation model outputs. After filtering out outliers, if this discrepancy exceeds a threshold, the ground station performs efficient LoRA fine-tuning. After N satellite updates, accumulated knowledge is transferred back to the foundation model using adaptive knowledge distillation. This adjusts distillation intensity based on satellite model accuracy, enabling robust many-to-one knowledge integration under bandwidth constraints. In summary, SGCL integrates on-orbit OOD filtering, selective transmission, update trigger, parameter-efficient LoRA updates, and adaptive distillation to enable continuous model evolution despite extreme bandwidth limitations.

We present the SGCL benchmark (SGCL-Bench) for simulating multisatellite on-orbit task execution and SGCL. The benchmark uniquely integrates multisource datasets with task-specific annotation strategies, leveraging GEOBench-VLM and VRSBench for conventional foundation tasks (high sample), while introducing seven specialized OOD tasks (low sample) to rigorously evaluate robustness under distribution shifts. SGCL-Bench simulates a realistic seven-satellite stream-processing environment where each satellite executes cyclical task workflows (5 tasks $\times 3$ cycles) reflecting orbital periodicity. We conducted sufficient experiments upon this benchmark to prove the superiority of our proposed SGCL framework.

II. Related Work

A. End–Cloud Collaborative Learning

Satellite–Earth collaborative learning can be seen as a variant of end–cloud collaborative learning, which refers to a dynamic and adaptive process where end models and cloud models evolve bidirectionally to improve their capabilities and performance. Ding et al. [8] propose a collaborative framework where CloudCNN provides soft supervision to local EdgeCNNs for end training, while EdgeCNN handles real-time inference with vision input. DCCL [9] introduces an end–cloud collaboration framework specifically tailored for recommendation systems. COLLA [10] is a collaborative learning method for user location prediction. COLLA features personalized models tailored for each device while facilitating collective learning between the cloud and end devices. The cloud model serves as a global aggregator, distilling knowledge from multiple end models. Alibaba Luoxi Platform [11] proposes a slow-fast learning mechanism for end–cloud collaborative recommendation. MPDA [12] is a device–cloud collaborative learning framework that enhances on-device training by retrieving similar data from the cloud's global pool to augment the user's local data. CLUE [13] introduces a framework that enhances knowledge transfer in neural networks for edge computing environments, utilizing a multimodel boosting approach to dynamically extract and leverage significant parameters from one node to boost another's predictive performance.

Existing end–cloud collaborative learning frameworks assume that edge devices possess training capability. For instance, FedRSCLIP [14] is a distributed learning framework specifically designed for remote sensing applications, but it similarly requires client devices to perform local training. However, in SGCL scenarios, orbital training is currently impractical due to resource constraints and the higher priority of primary satellite tasks. This raises the critical question of determining the criteria for updating satellite models. Furthermore, while existing studies highlight the difficulty of fusing knowledge from multiple edge models into a central model and recognize the advantages of "model-over-models" distillation over traditional "model-over-data" methods, a simplistic averaging of KL divergence between output distributions for knowledge aggregation risks degrading central model performance. This is particularly problematic if certain edge models exhibit significantly lower performance.

B. Remote Sensing VLMs

Vision-language models (VLMs) are defined as a family of AI models that integrate computer vision and natural language processing to achieve a holistic multimodal understanding of visual and textual information [15]. Recently, large VLMs (LVLMs) have emerged as a forefront research direction in remote sensing (RS). By combining visual comprehension, human-like world knowledge, and advanced reasoning capabilities, LVLMs significantly enhance the analysis of RS data in an interpretable and user-friendly manner [5]. Consequently, we select LVLMs as the base architecture for validating our proposed SGCL framework.

Concretely, initial research focused on adapting generic VLMs to RS tasks. Building upon foundational RS-VQA frameworks proposed in [16], which demonstrated that language-guided queries enhance scene interpretation accuracy, subsequent studies addressed domain-specific challenges. Chappuis et al. [17] identified and mitigated language biases in spatial attribute recognition. Hackel et al. [18] introduced a lightweight transformer architecture to improve computational efficiency for on-device deployment. To solve the limitations of low-level features, reliance on annotated data for fine-tuning, and lack of language understanding for zero-shot tasks, the first purpose-built vision-language foundation model for RS, RemoteCLIP [19], was proposed.

Recently, RSGPT [20] is the first RS LVLM designed for remote sensing image captioning and VQA. Leveraging high-quality human annotations and a novel cross-modal attention mechanism, it achieves fine-grained feature fusion for these tasks. EarthGPT [21] integrated SAR and optical imagery with textual metadata for multisensor Earth observation tasks, while SkyeyeGPT [22] specialized in high-resolution aerial image analysis through a spatiotemporal fusion module, enabling dynamic scene evolution tracking. GeoChat [23] introduced a conversational RS-specific VLM with multitasking capabilities (e.g., image captioning, change detection, and object counting), leveraging a hybrid encoder–decoder architecture to unify geospatial dialog. Meanwhile, SkySense [24] established

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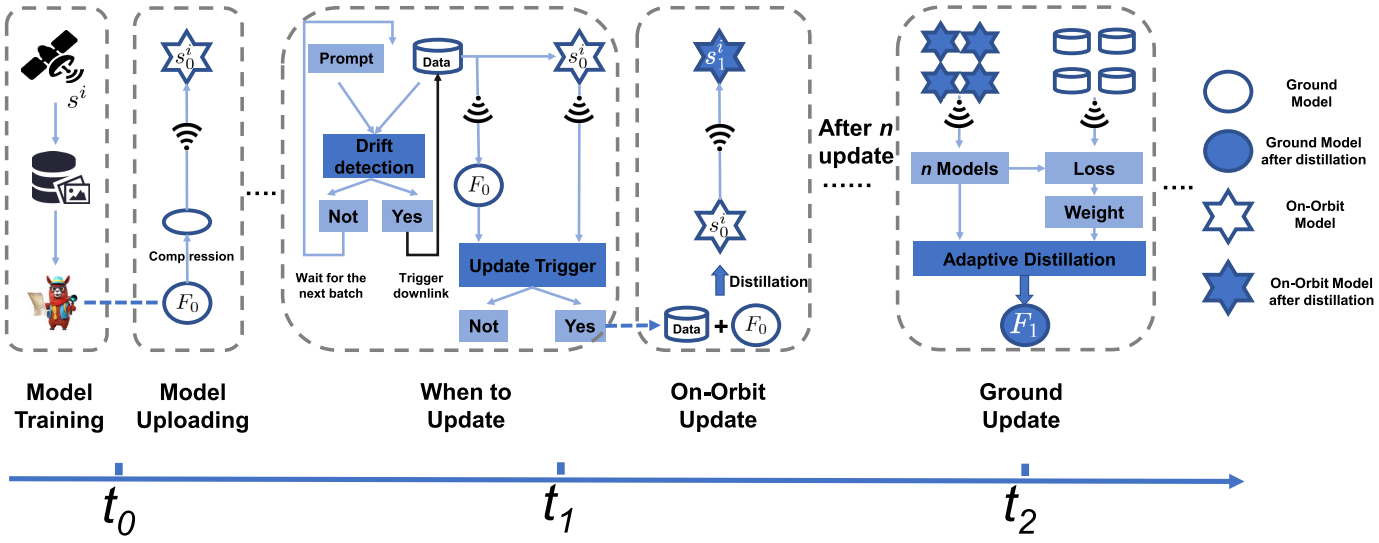


Fig. 2. Learning flow of SGCL framework.

itself as the largest multimodal RSFM to date, pretrained on 21.5 million temporal sequences of optical and SAR data using a novel spatiotemporal encoder. It demonstrated superior generalization, outperforming existing models like GFM, SatLas, and Scale-MAE across diverse tasks and benchmarks. Subsequently, SkySense++ [5] pioneered a semantic-enhanced multimodal foundation model for Earth observation, featuring factorized spatiotemporal encoders and progressive pretraining to achieve superior few-shot performance across 12 tasks in 7 domains. While these pioneering foundation models (such as RemoteCLIP [19], GeoRSCLIP [25], and GeoChat [23]) provide exceptional baseline cognitive capabilities leveraging massive image–text pairs, deploying them directly on satellites remains computationally prohibitive. Our SGCL framework acts as a complementary system-level solution, ensuring that the knowledge from such powerful ground-based models can be efficiently compressed, transferred, and continuously updated on-orbit.

III. Satellite–Ground Collaborative Learning

The SGCL framework consists of a set of satellites and the ground station. We illustrate the learning process using a star topology, where each satellite is connected to the ground station. In SGCL, the ground station trains and maintains a foundation model F, while the ith satellite has its own customized on-orbit model s^i.

A. Learning Flow

As shown in Fig. 2, the SGCL framework consists of the following six stages.

Stage A (Bootstrapping): The ground station trains the first foundation model from the initial dataset, such as a remote sensing vision-language model. At this stage, there are already massive volumes of Earth system data, numerous datasets, and open-source foundation models available to complete the construction.

Stage B (First Pulling): When the satellite joins the SGCL framework at time t_0, it requests the latest model from the ground station (or complete the latest model loading before satellite launch). The ground station compresses its latest foundation model F_0 into a small one and sends it to satellite s^i via the uplink. s^i uses the compressed small model as its first on-orbit model s_0^i to perform inference in the following time period $t = t_1 - t_0$.

Stage C (OOD Detector): To address distribution shifts and OOD inputs during on-orbit inference, we introduce a dual-mechanism data-shift monitoring module. This system detects both significant input deviations from the training distribution and novel prompts not encountered during ground-based training. For a batch $\{(I_k, T_k)\}_{k=1}^{B}$ at time t, the satellite extracts visual features and text features using its on-orbit model encoders. A reconstruction error is computed via the autoencoder trained on initial features, where deviations $\mathcal{RE}_k > \tau_{\text{RE}}$ flag visual anomalies. Concurrently, we employ the Mahalanobis distance to evaluate prompt shifts and cosine similarity to evaluate the multimodal misalignment. Based on the three indicators mentioned above, calculate the anomaly rate p. If $p > \theta_p$, send a data-shift alert to the ground station and proceed to Stage D.

Stage D (Update Trigger): We propose an update trigger that automatically assesses whether the on-orbit model struggles to handle changing scenarios and tasks, based on the discrepancies between the on-orbit model results and the foundation model results. Section III-D will describe the update trigger in detail.

Stage E (On-Orbit Model Update): If Stage D determines that an update is necessary (at time t_2), the ground station will perform LoRA fine-tuning on the on-orbit model s_1^i based on the latest foundation model F_1 through knowledge distillation. The purpose of LoRA fine-tuning is to significantly reduce the parameters transmitted during each model update while avoiding the forgetting of essential context embedded in the fully tuned model. To meet the varying needs of on-orbit tasks, the update process can be either manually assisted or fully automated. In the automated mode, the results of the foundation model for the bth batch of data are used as pseudo-labels, and the ground model serves as the teacher

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model to fine-tune the on-orbit model using these pseudo-labels. In the manually assisted mode, experts correct the pseudo-labels for the bth batch of data to achieve better on-orbit model performance. Thus, the on-orbit model can learn from both its local data and the ground model. Section III-E will describe the knowledge distillation process in detail. Once the on-orbit model s_2^i is generated, its parameters are uploaded to the satellite for the update.

Stage F (Foundation Model Update): After N on-orbit model updates, the ground station will also update its foundation model. Given that this update involves many-to-one knowledge transfer from smaller models to a larger model, we employ adaptive knowledge distillation. The distillation intensity is self-adaptively controlled based on the correctness of the on-orbit models' predictions. For more details, please refer to Section III-F. This stage results in a new foundation model.

The on-orbit model and the ground model will be updated asynchronously continuously. For example, at time t_2, the station employs the latest foundation model F_2 and distills the knowledge into on-orbit model s_1^i using the bth batch of data, resulting in a new on-orbit model s_2^i, and then pushes it to the satellite. Similarly, the station updates the foundation model from time to time using N updated on-orbit models. We use adaptive distillation when transferring knowledge from multiple on-orbit models to a single ground foundation model and perform vanilla distillation when transferring knowledge from the ground foundation model to on-orbit models separately.

B. Base Model

As discussed in Section II, we focus on remote sensing VLMs. Since the only project that currently has training code and model weights is GEOchat [23], we chose it as our ground foundation model. GeoChat demonstrates prior zero-shot performance on various remote sensing tasks such as visual question answering, classification, image captioning, and visually grounded conversations. It follows the architecture of LLaVA-v1.5 [26], which comprises three core modules as follows.

1) *Visual Encoder*: A CLIP-ViT-L/14 model [27], pretrained on massive image–text pairs, responsible for extracting robust semantic visual features from high-resolution remote sensing imagery.
2) *Cross-Modal Projector*: A two-layer multilayer perceptron that maps the high-dimensional visual tokens into the embedding space of the language model, ensuring strict modality alignment.
3) *Large Language Model*: The Vicuna-v1.5 (7B parameters) [28] serves as the central cognitive engine, performing complex visual reasoning and generating textual responses based on the projected visual features and user prompts.

Conversely, to strictly adhere to the severe computational and memory constraints of satellite platforms, the on-orbit model utilizes **Qwen2-VL-2B** [29], a highly efficient and lightweight VLM. Qwen2-VL offers superior performance on visual-language tasks with significantly fewer parameters (only 2B). During the proposed SGCL process, the on-orbit model's representations are dynamically and efficiently updated utilizing low-rank adaptation (LoRA) [30] modules, thereby minimizing bandwidth overhead during uplink transmissions.

C. OOD Detector

To address distribution shifts and OOD inputs during on-orbit inference, we introduce a multimetric data-shift monitoring module. It is important to note that our framework employs a unified VLM paradigm. Under this architecture, even standard vision tasks are formulated as instruction-following processes utilizing default task templates. Consequently, baseline text embeddings are continuously generated and remain available for consistency evaluation across all task types.

The OOD detector does not rely on a single metric but operates as an integrated ensemble where each component is strictly assigned to a distinct type of distribution shift under satellite resource constraints. For a batch $\{(I_k, T_k)\}_{k=1}^{B}$ processed by satellite s^i, the vision encoder E_v outputs visual features $v_k \in \mathbb{R}^{d_v}$, while the text encoder E_t generates prompt embeddings $t_k \in \mathbb{R}^{d_t}$.

First, to target visual anomalies, a lightweight autoencoder AE : $\mathbb{R}^{d_v} \to \mathbb{R}^{d_v}$ computes the reconstruction error $\mathcal{RE}_k = \|\mathrm{AE}(v_k) - v_k\|_2^2$. It was pretrained during Stage-B. Visual anomalies are flagged when $\mathcal{RE}_k$ exceeds the dynamic threshold $\tau_{\mathrm{RE}} = \mu_{\mathrm{RE}}^{\mathrm{train}} + 3\sigma_{\mathrm{RE}}^{\mathrm{train}}$, where $\mu_{\mathrm{RE}}^{\mathrm{train}}$ and $\sigma_{\mathrm{RE}}^{\mathrm{train}}$ are training-set moments.

Second, serving as our primary defense against Format OOD, a Mahalanobis distance-based OOD module operates on text features to capture structural or syntactic deviations: $\mathcal{O}_k = (t_k - \mu_t)^\top \Sigma_t^{-1}(t_k - \mu_t)$, where $\mu_t \in \mathbb{R}^{d_t}$ and $\Sigma_t \in \mathbb{R}^{d_t \times d_t}$ are the training-set feature mean and covariance matrix. $\mathcal{O}_k > \tau_O$ is regarded as the criterion for OOD, where τ_O is the 95% value in $\mathcal{O}_k^{\mathrm{train}}$. This metric successfully catches linguistic anomalies even when the underlying semantic similarity remains deceptively high.

Third, acting as a dedicated safety net for severe cross-modal semantic misalignment, multimodal consistency is assessed through cosine similarity $\mathcal{S}_k = v_k \cdot t_k / \|v_k\| \|t_k\|$. Because VLMs utilize aligned embedding spaces, this metric reliably triggers an alert when a text instruction queries objects entirely absent from the mutated visual scene. Anomalies are triggered by $\mathcal{S}_k < \tau_S$, where $\tau_S = \mu_S^{\mathrm{train}} - 3\sigma_S^{\mathrm{train}}$.

The shift severity $p = 1/B \sum_{k=1}^{B} \mathbb{I}[\mathcal{RE}_k > \tau_{\mathrm{RE}} \vee \mathcal{O}_k > \tau_O \vee \mathcal{S}_k < \tau_S]$ is evaluated against a threshold θ_p, and upon breach ($p > \theta_p$), an alert is transmitted to Stage D.

In addition, the OOD detection module incorporates a deduplication mechanism that suppresses data transmission for recurrent anomalies. When a task's feature fingerprint exhibits similarity of γ to previous transmissions within several tasks and has triggered at least two downlinks, only logging is performed. The joint fingerprint $H_{joint} = H_v \oplus H_t$ combines visual and textual features via $H_v = \mathrm{SimHash}(\mathrm{PCA}(v_k, 32))$ and $H_t = \mathrm{MD5}(\arg\max_c p^s(y_c|t_k))$. This design prevents redundant downlinks while avoiding model oscillation caused by conflicting LoRA updates that exacerbate catastrophic forgetting. By

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maintaining a sliding-window cache of fingerprint, timestamp, and count tuples, the system redirects computational resources toward novel distribution shifts.

D. Update Trigger

For a batch of data $\{x_j\}_{j=1}^{M}$, we calculate the prediction probability distribution $q^F(y|x_j)$ and prediction confidence $C_g(x_j)$ from the ground foundation model, and the prediction probability distribution $p^s(y|x_j)$ from the satellite on-orbit model. For each data point, we calculate the Kullback–Leibler (KL) divergence between the two models' prediction probability distributions

$$\mathcal{D}_{\mathrm{KL}}\left(q^F\left(y|x_j\right) \parallel p^s\left(y|x_j\right)\right) = \sum_y q^F\left(y|x_j\right) \log\left(\frac{q^F\left(y|x_j\right)}{p^s\left(y|x_j\right)}\right). \tag{1}$$

Then, we use the prediction confidence of the ground foundation model to weight the KL divergence, obtaining the weighted discrepancy for each data point

$$d_j = C_g\left(x_j\right) \times \mathcal{D}_{\mathrm{KL}}\left(q^F\left(y|x_j\right) \parallel p^s\left(y|x_j\right)\right). \tag{2}$$

Higher confidence indicates that the ground model's prediction is more reliable, and the corresponding discrepancy weight should be larger. Before averaging the set of weighted discrepancies $\{d_j\}_{j=1}^{M}$, it is necessary to detect and filter outliers to avoid their misleading impact on the module's decision-making. We calculate the median $\tilde{d}$ and median absolute deviation (MAD) of the weighted discrepancies and define a threshold to identify outliers in the discrepancies

$$\theta = \tilde{d} + k \times \mathrm{MAD} \tag{3}$$

where k is the threshold coefficient, commonly set to $k = 3$. We treat data points with weighted discrepancies exceeding the threshold θ as outliers and filter them out to obtain the filtered discrepancy set D'. Finally, we compute the average of the filtered weighted discrepancies to obtain the mean weighted discrepancy $\bar{d}$ and trigger the model update when the $\bar{d}$ exceeds the predefined threshold δ.

E. On-Orbit Model Update

On the satellite side, knowledge is transferred from the ground foundation model to the on-orbit model. The foundation model acts as a teacher, and we perform knowledge distillation on the bth batch of data. The loss function of the on-orbit model i can be formulated as

$$\mathcal{L}_i^s = \lambda_s \mathcal{L}_{\mathrm{CE}}\left(y_i, p_i^s\right) + \left(1 - \lambda_s\right) \mathcal{D}_{\mathrm{KL}}\left(q^F, p_i^s\right) \tag{4}$$

where $\mathcal{L}_{\mathrm{CE}}$ denotes the cross entropy; $\mathcal{D}_{\mathrm{KL}}$ denotes the KL divergence; and superscripts F and s denote the foundation model and the on-orbit model, respectively. Here, y_i represents the label of the data, while p_i^s and q^F represent the output probabilities of the on-orbit model and foundation model, respectively. We employ early stopping to avoid overfitting on limited data. As mentioned earlier, SGCL has both annotation and automated settings, i.e., here y_i is either manually labeled or pseudo-labeled derived from the foundation model F.

F. Foundation Model Update

After N on-orbit model updates, we employ all on-orbit models to teach the foundation model by fine-tuning it on the K batches of data. Previous methods aim to match the output probabilities of the cloud model to the average of the output probabilities of each client model, which ignores the variations between client models (on-orbit models). We dynamically allocate distillation weights based on the individual on-orbit model's performance. In the "annotation" setting, we have manual annotations for data and the weights γ_i of on-orbit model n are calculated as follows:

$$\gamma_i = \frac{1}{N-1}\left(1 - \mathrm{Softmax}\left(\mathcal{L}_{\mathrm{CE}}\left(y_i, p_i^s\right)\right)\right) \tag{5}$$

where $\mathcal{L}_{\mathrm{CE}}(y_i, p_i^s)$ represents the cross-entropy loss of client i on their coresponding batch of data. Next, the weights are dynamically assigned to different local models

$$\mathcal{L}^F = \lambda_f \mathcal{L}_{\mathrm{CE}}\left(Y, q^F\right) + \left(1 - \lambda_f\right) \frac{1}{N} \sum_{i=1}^{N} \gamma_i \mathcal{D}_{\mathrm{KL}}\left(p_i^s, q^F\right) \tag{6}$$

where Y denotes the labels of all the data collected by the N updated on-orbit models and q^F and p_i^s denote the class probabilities of the base model and the on-orbit model i for the full data, respectively.

Under the "automated" setting, we have no annotation of the data and therefore use model output consistency to assign weights. Specifically, for the entire collected data, we calculate the inverse of the Jensen–Shannon divergence to determine the average similarity between the output of on-orbit model i and all other on-orbit models. By normalizing this similarity, we obtain the distillation weight for model i. In addition, due to the absence of data labeling, the cross-entropy loss function in (6) is also omitted in the fully automated setting.

G. Theoretical Analysis of the SGCL Framework

To rigorously characterize the learning dynamics within the SGCL framework, we formulate the collaborative process as a constrained empirical risk minimization (ERM) problem. Let $\mathcal{R}_t(s^i) = \mathbb{E}_{(x,y)\sim\mathcal{D}_t}[\mathcal{L}(s^i(x), y)]$ denote the expected risk of the on-orbit model s^i on the dynamic data distribution $\mathcal{D}_t$ at time t. Due to continuous spatial–temporal distribution shifts, the local empirical risk $\hat{\mathcal{R}}_t(s^i)$ inevitably increases.

Our update trigger acts as a structural constraint by monitoring the confidence-weighted KL divergence. By enforcing the condition $\bar{d} \le \delta$, the framework established a functional trust region in the output distribution space. This mathematically bounds the maximum allowable divergence between the satellite's local predictive distribution p^s and the foundation model's robust prior q^F, ensuring that the on-orbit model does not deviate into unreliable high-entropy states during minor fluctuations.

When a significant OOD event causes the discrepancy to exceed the bound δ, the LoRA fine-tuning process is initiated. From an optimization perspective, the distillation loss $\mathcal{L}_i^s$ [see (4)] functions as a regularized projection. It projects the high-capacity representational knowledge of the foundation model F into the low-rank subspace of the satellite model. In this

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TABLE I
DATA SOURCES AND ANNOTATION METHODOLOGIES FOR SGCL-BENCHMARK

Task Name	Data Source	Annotation
Agricultural Anomaly Detection	Agriculture-Vision [31]	Using Agriculture-Vision's anomaly type annotations, we generated QA pairs via GPT-4O, followed by expert validation
Disaster Caption	xBD Dataset [32]	Disaster experts drafted preliminary situational reports based on XBD dataset annotations, which were then linguistically refined through GPT-4O before final expert confirmation
Relation Reasoning-Disaster	xBD Dataset [32]	Disaster experts drafted preliminary situational reports based on XBD dataset annotations, which were then linguistically refined through GPT-4O before final expert confirmation
VQA-formal shift	VRSBench [33]	Utilizing GPT-4O, we generated datasets by preserving task structures and ground-truth answers while systematically introducing format perturbations (e.g., complex syntax, colloquial expressions, spurious logic, rare terms, or unconventional decomposition). Each output was professionally examined for task integrity, valid shifts, answer consistency, and grammar
Ship Type Classification	GEOBench-VLM [34]	Original annotations
Aircraft Type Classification	GEOBench-VLM [34]	Original annotations
Land Use Classification	GEOBench-VLM [34]	Original annotations
Spatial Relation reasoning	GEOBench-VLM [34]	Original annotations
Scene Classification	GEOBench-VLM [34]	Original annotations
Image Caption	VRSBench [33]	Original annotations
Visual Question Answering (VQA)	VRSBench [33]	Original annotations

process, the KL-divergence term acts as a functional regularizer, which prevents catastrophic forgetting by ensuring that the updated parameters s_{t+1}^i remain close to the generalized knowledge manifold defined by the foundation model.

Furthermore, during the foundation model update stage (Stage F), the adaptive weighting mechanism γ_i addresses the challenge of distributional heterogeneity. Unlike traditional federated averaging (FedAvg), which assigns uniform weights, our method calculates γ_i based on local empirical loss (5). This transforms the global update into a quality-aware aggregation problem

$$\mathcal{L}^F = \lambda_f \mathcal{L}_{\mathrm{CE}}\left(Y, q^F\right) + \left(1-\lambda_f\right)\frac{1}{N}\sum_{i=1}^{N}\gamma_i \mathcal{D}_{\mathrm{KL}}\left(p_i^s \,\|\, q^F\right). \quad (7)$$

This formulation theoretically guarantees that the foundation model selectively aggregates high-fidelity knowledge from reliable on-orbit distributions while discounting "noisy" updates from satellites encountering extreme anomalies, thereby ensuring stable convergence in the many-to-one knowledge transfer process.

IV. SGCL-BENCHMARK

We developed the SGCL benchmark to simulate multisatellite on-orbit task execution and SGCL.

A. Dataset Source

As shown in Table I, four tasks (visual question answering, scene classification, image caption, and spatial relation classification) leverage established GEOBench-VLM [34] and VRSBench [33] datasets with their original annotations. These constitute the majority of samples and represent conventional scenarios (e.g., VQA includes category, presence, quantity, color, shape, size, position, direction, scene, and reasoning tasks). Furthermore, SGCL-benchmark (SGCL-bench) introduces dedicated tasks as OOD test cases to assess model robustness.

For *domain OOD*, we implemented an agricultural anomaly detection task using the Agriculture-Vision dataset [31]. We use GPT-4O to generate the question-and-answer pairs, and then conduct expert verification. Disaster caption and spatial relation-disaster task using XBD data [32], where disaster experts drafted reports polished by GPT-4O and finalized by experts. These test model responses to abrupt surface changes (disasters) or under-represented agricultural contexts.

For *fine-grained OOD*, we utilized GEOBench-VLM [34]'s original annotations for ship, aircraft, and land use classification. For *format OOD*, we employed structured standard QA pairs from the VRSBench dataset as input to ensure task definitions are clear and answers are unambiguous. Subsequently, the GPT-4O interface is invoked for batch rewriting, with its system prompt requiring strict constraints on the generation logic: it must preserve the original task type and answer while mandatorily introducing at least one format OOD deviation feature—such as complex nested sentence structures, colloquial expressions, misleading logical conditions, substitution with obscure terminology, or unconventional task decomposition workflows. Diversity in generation is controlled via the temperature parameter. For each batch, ten standard instructions are inputted, and GPT-4O outputs variant proposals in the format ["raw-prompt":..., "OOD-prompt":...]. The generated outputs undergo professional format filtering, specifically verifying four fundamental rules: whether variants maintain the original task type, introduce valid deviations, preserve answer consistency, and contain no grammatical errors. This dataset design explicitly serves as a controlled

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Algorithm 1 Satellite–Ground Collaborative Learning

Input: Initial dataset $\mathcal{D}_{\text{init}}$, Satellites $S = \{s^1, \ldots, s^N\}$, OOD thresholds $\tau_{\text{RE}}, \tau_O, \tau_S$, Update thresholds θ_p, δ.
Output: Updated ground model F_T, Updated on-orbit models $\{s_T^i\}_{i=1}^N$.
Stage A: Bootstrapping
Train foundation model: $F_0 \leftarrow \text{Bootstrap}(\mathcal{D}_{\text{init}})$
Stage B: First Pulling
for each satellite $i \in [1, N]$ **do**
  Compress foundation model: $F_0^{\text{comp}} \leftarrow \text{Compress}(F_0)$
  Initialize on-orbit model: $s_0^i \leftarrow F_0^{\text{comp}}$
end for
Main Loop:
for time step $t = t_1, t_2, \ldots$ **do**
  Stage C: OOD Detector (per satellite i)
  for batch $\{(I_k, T_k)\}_{k=1}^B$ **do**
    $v_k \leftarrow E_v(I_k)$, $t_k \leftarrow E_t(T_k)$
    $\mathcal{RE}_k \leftarrow \|\text{AE}(v_k) - v_k\|^2$
    $\mathcal{O}_k \leftarrow (t_k - \mu_t)^\top \Sigma_t^{-1} (t_k - \mu_t)$
    $\mathcal{S}_k \leftarrow \frac{v_k \cdot t_k}{\|v_k\| \|t_k\|}$
    $p \leftarrow \frac{1}{B} \sum_{k=1}^B \mathbb{I}[\mathcal{RE}_k > \tau_{\text{RE}} \vee \mathcal{O}_k > \tau_O \vee \mathcal{S}_k < \tau_S]$
    if $p > \theta_p$ **then**
      Send alert to ground station
    end if
  end for
  if alert received **then**
    Stage D: Update Trigger
    Compute discrepancies: $d_j = C_g(x_j) \cdot \mathcal{D}_{\text{KL}}(q^F(y|x_j) \parallel p^s(y|x_j))$
    $\bar{d} \leftarrow \text{FilteredMean}(\{d_j\}, \theta = 3)$
    if $\bar{d} > \delta$ **then**
      Stage E: On-Orbit Model Update
      $\mathcal{L}_i^s \leftarrow \lambda_s \mathcal{L}_{\text{CE}}(y_i, p_i^s) + (1 - \lambda_s) \mathcal{D}_{\text{KL}}(q^F \parallel p_i^s)$
      Update model: $s_{t+1}^i \leftarrow \text{LORA-Finetune}(s_t^i, \mathcal{L}_i^s)$
    end if
  end if
  if N updates completed **then**
    Stage F: Foundation Model Update
    Calculate weights: $\gamma_i \leftarrow \frac{1}{N-1}(1 - \text{Softmax}(\mathcal{L}_{\text{CE}}(y_i, p_i^s)))$
    $\mathcal{L}^F \leftarrow \lambda_f \mathcal{L}_{\text{CE}}(Y, q^F) + (1 - \lambda_f) \sum_{i=1}^N \gamma_i \mathcal{D}_{\text{KL}}(p_i^s \parallel q^F)$
    Update model: $F_{t+1} \leftarrow \text{LORA-Finetune}(F_t, \mathcal{L}^F)$
  end if
end for=0

experiment to verify and quantify how prompt variations, such as syntax shifts and colloquialisms, affect the outputs of on-orbit VLMs. Note that GPT-4O functions solely as a linguistic form converter and text polisher in this workflow. We do not use GPT-4O to synthesize artificial remote sensing scenarios or hallucinate ground-truth facts.

These seven specialized OOD tasks (contrasted with the high-sample conventional tasks) represent significantly smaller proportions within the training set. This deliberately skewed distribution explicitly positions them as OOD evaluation scenarios under real-world distribution shifts.

B. Simulating Satellite Task Execution

Based on the previous data collection and annotation work, we constructed a multisatellite dataset simulating the on-orbit task flow. The test data for each satellite do not overlap with each other. Considering typical LEO edge computing simulations using three [35] or five [36] satellites, we configured a seven-satellite stream-processing benchmark. This design enables flexible validation for subsequent researchers testing fleets ranging from one to seven satellites.

Accounting for orbital periodicity and task diversity, each satellite performs five distinct tasks cyclically three times (totaling 15 batches per satellite). Detailed task allocation for each satellite is provided in Table II.

C. Dataset Partitioning

Since GEOBench-VLM was initially designed solely for evaluation purposes, we adopted a 90/10 test/train split for both its dataset and our self-annotated samples covering disaster caption, disaster relational reasoning, agricultural anomaly detection, and format OOD tasks. This promotes model adaptation to specific tasks and ensures fair evaluation. Dataset splits align with VRSBench's methodology. Table II shows the number of samples in the test set.

V. Experiments

A. Implementation Details

Our ground foundation model adopts the GeoChat architecture [23], initialized with the pretrained CLIP-ViT-L/14 visual encoder and Vicuna-v1.5-7B language model [28]. We employ LoRA [30] with rank $r = 64$ on the attention weight matrices (W_q, W_k, and W_v) of the LLM to adapt the model efficiently. All training is performed at an image resolution of 504×504. The model is trained using multimodal instruction templates across diverse vision-language tasks with the AdamW optimizer (global batch size 144) and a cosine learning rate scheduler. To ensure high-quality knowledge transfer during fine-tuning, we tailor the label validation process based on the operational mode. In the annotated mode, we integrate a human-in-the-loop verification step where domain experts use a ground-based dashboard to inspect and correct semantic or formatting errors in the foundation model's pseudo-labels before updating. In the automated mode, we apply a confidence-based filtering mechanism. Furthermore, to guarantee training stability and prevent catastrophic forgetting on these small OOD batches, the fine-tuning process is restricted to a maximum of ten epochs, incorporating an early stopping mechanism with a patience of three epochs based on training loss convergence.

To meet stringent satellite computational constraints, we transfer the foundation model's multimodal capabilities to the lightweight Qwen2-VL-2B architecture [29] via progressive knowledge distillation. Synchronously optimize visual–text alignment using mean squared error (mse) loss between foundation model (v_k^F) and satellite model (v_k^s) embeddings: $\mathcal{L}_{\text{align}} = 1/B \sum_{k=1}^B \|v_k^F - v_k^s\|^2$ on Geochat-generated pseudo-labels. Apply task-specific LoRA fine-tuning (rank $r = 64$ on W_q, W_k, and W_v) using KL divergence distillation loss

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TABLE II

DISTRIBUTION OF TASK FLOWS EXECUTED BY EACH SATELLITE AND THE NUMBER OF SAMPLES (TEST SET) FOR EACH TASK IN THE SGCL-BENCH. "AAD" DENOTES AGRICULTURAL ANOMALY DETECTIO AND "SRD" MEANS SPATIAL RELATION-DISASTER

Satellite	Cycle-1					Cycle-2					Cycle-3				
A	Caption	Disaster Caption	VQA	Spatial Relation	VQA-formal shift	Caption	Disaster Caption	VQA	Spatial Relation	VQA-formal shift	Caption	Disaster Caption	VQA	Spatial Relation	VQA-formal shift
	519	50	1089	35	50	519	50	1089	35	50	519	50	1089	35	50
B	AAD	Spatial Relation	Land Use	VQA	Caption	AAD	Spatial Relation	Land Use	VQA	Caption	AAD	Spatial Relation	Land Use	VQA	Caption
	40	35	41	1089	519	40	35	41	1089	519	40	35	41	1089	519
C	VQA	Caption	Scene Class	Disaster Caption	Ship Type	VQA	Caption	Scene Class	Disaster Caption	Ship Type	VQA	Caption	Scene Class	Disaster Caption	Ship Type
	1089	519	41	50	34	1089	519	41	50	35	1088	520	41	50	35
D	Spatial Relation	SRD	Land use	VQA-formal shift	VQA	Spatial Relation	SRD	Land use	VQA-formal shift	VQA	Spatial Relation	SRD	Land use	VQA-formal shift	VQA
	35	35	41	50	1088	35	35	41	50	1088	35	35	41	50	1088
E	VQA	VQA-formal shift Aircraft Type		Spatial Relation	Caption	VQA	VQA-formal shift Aircraft Type		Spatial Relation	Caption	VQA	VQA-formal shift Aircraft Type		Spatial Relation	Caption
	1089	50	30	35	519	1089	50	30	35	519	1089	50	30	35	520
F	Ship Type	Caption	Scene class	VQA-formal shift	AAD	Ship Type	Caption	Scene class	VQA-formal shift	AAD	Ship Type	Caption	Scene class	VQA-formal shift	AAD
	34	519	41	50	40	34	520	41	50	40	35	520	41	50	40
G	SRD	Caption	Land use	Scene class	VQA	SRD	Caption	Land use	Scene class	VQA	SRD	Caption	Land use	Scene class	VQA
	35	519	41	41	1089	35	519	41	41	1089	35	519	41	41	1089

TABLE III

PERFORMANCE FOR EACH SATELLITE ON EACH MISSION IN A SEVEN-SATELLITE SETUP, WHERE GREEN BACKGROUND MEANS THAT ON-ORBIT MODEL UPDATE EXECUTED ON THAT BATCH OF DATA, GRAY BACKGROUND INDICATES THAT DATA DOWNLINK WAS TRIGGERED BY DRIFT DETECTOR, BUT NO UPDATE WAS TRIGGERED, † MEANS THAT THE GROUND FOUNDATION MODEL UPDATE WAS PERFORMED, "AVG" MEANS AVERAGE, "CLS" MEANS CLASSIFICATION, AND "CAP" MEANS CAPTION

No.	Setting	Cycle-1(%)					Cycle-2(%)					Cycle-3(%)					AVG(CLS)	AVG(CAP)
Satellite A	without SGCL	29.7	25.7	74.5	74.2	46.0	29.8	25.1	73.3	77.1	44.0	30.2	25.5	72.5	77.1	48.0	65.2	27.6
	with SGCL	29.7	25.7	74.6	77.1	46.0 †	30.1	28.2	73.4	74.2	64.0	30.5	28.1	72.7	80.0	80.0	71.6	28.7
Satellite B	without SGCL	37.5	71.4	51.2	72.2	30.1	40.0	68.5	53.6	71.9	29.4	35.0	71.4	53.6	71.8	29.8	58.2	29.7
	with SGCL	37.5	71.4	51.2†	72.3	30.0	57.5	68.5	63.4	71.9	29.4	70.0	71.4	65.8	72.7	29.8	64.4	29.7
Satellite C	without SGCL	72.6	29.9	80.4	24.9	32.3	73.3	29.6	80.4	23.0	37.1	73.6	29.8	82.9	24.8	25.7	62.8	27.0
	with SGCL	72.6	29.9	80.4	24.9	32.3	73.2	29.8	80.4	25.1	48.5	73.7	30.3	82.9	26.7	48.5	65.8	27.7
Satellite D	without SGCL	68.5	37.1	56.0	54.0	70.1	71.4	40.0	60.9	62.0	71.5	68.5	45.7	58.5	60.0	73.6	59.8	-
	with SGCL	68.5	37.1	56.0	54.0	70.5	74.2	54.2	68.2	72.0	71.8	74.2	65.7	70.7	76.0	74.5	65.8	-
Satellite E	without SGCL	71.9	52.0	43.3	71.4	29.6	72.2	54.0	40.0	77.1	29.9	74.1	52.0	46.6	71.4	30.0	60.53	29.8
	with SGCL	71.9	52.0	43.3	71.4	29.8	72.1	66.0†	50.0	77.1	29.9	73.9	76.0	53.3	74.2	29.9	65.1	29.8
Satellite F	without SGCL	38.2	29.5	82.9	50.0	30.0	44.1	29.4	82.9	44.0	35.0	42.8	30.2	85.3	48.0	35.0	51.5	29.7
	with SGCL	38.2	29.5	82.9	50.0	32.5	55.8	29.3	85.3	64.0	47.5†	54.2	30.3	85.3	82.0	52.5	60.88	29.7
Satellite G	without SGCL	34.2	29.4	53.6	82.9	71.3	37.1	29.2	48.7	85.3	72.6	37.1	29.1	53.6	87.8	71.6	61.3	29.2
	with SGCL	34.2	29.4	53.6	85.3	71.4	57.1	29.2	58.5	85.3	72.7	71.4	29.2	65.8	90.2	72.3	68.2	29.2

$\mathcal{L}_{\text{distill}} = \mathcal{D}_{\text{KL}}(q^F \| p^s)$. Optimization uses AdamW ($\beta_1 = 0.9$ and $\beta_2 = 0.98$) with cosine annealing (base LR = $2e^{-5}$ and 0.05 warmup) and global batch size 144.

The hyperparameters for the OOD detector and update trigger are statistically derived from the training data distribution. Specifically, we use the 10% training split as a calibration set to determine the thresholds. For the visual reconstruction error and multimodal consistency metrics, we compute the mean (μ) and standard deviation (σ) on this calibration set and apply the 3σ rule (e.g., $\tau_{\text{RE}} = \mu_{\text{RE}}^{\text{train}} + 3\sigma_{\text{RE}}^{\text{train}}$). For prompt shift detection, the threshold τ_O is set to the 95th percentile of the Mahalanobis distance observed within the calibration set. The update trigger threshold θ is determined using the MAD of the discrepancy distribution: $\theta = \tilde{d} + 3 \times \text{MAD}$, where $\tilde{d}$ is the median discrepancy computed on the calibration set. Additionally, we set the anomaly ratio threshold $\theta_p = 0.1$ and the discount factor $\gamma = 0.9$ based on cross-validation performance on the training set to balance detection sensitivity and false alarm rate. The knowledge distillation balancing coefficients λ_s and λ_f are empirically set to 0.7 based on cross-validation performance on the training set, prioritizing on-orbit task accuracy while preserving foundational knowledge alignment. For more details, please refer to the code.

B. SGCL Versus "Learn Then Deploy"

We validated the overall effectiveness of the SGCL framework in comparison with the "learn then deploy" paradigm, where on-orbit models remain static without performance-based updates.

An initial simulation involved a seven-seven-satellite-annotation setting, where each satellite processed 15 tasks. Drift detection automatically triggered data downlinking during processing, while update detection subsequently activated asynchronous updates for satellite and ground models. The processing results are presented in Table III. The stepwise performance for each satellite in the configuration of three-satellite/one-satellite is detailed in Tables IV and V. To ensure experimental reliability, three repeated trials were conducted, with average results and standard deviations reported in Table VI SGCL achieved an average classification accuracy improvement of 6.2% over the baseline. Moreover, even when only three satellites participated, as demonstrated in Table VII.5, our method maintained a significant lead with a 4.7% enhancement in average classification accuracy.

For caption performance evaluation using ROUGE-L scores, in the SGCL-bench, the conventional caption task (non-OOD) occurs 18 times, while the disaster caption task (domain OOD) occurs only six times. Therefore, the improvement in the average caption performance was overwhelmed by the non-OOD tasks, and the performance of the disaster caption task significantly improved by 6.8% (as shown in Table VIII).

Additionally, recognizing the limitations of fixed metrics for evaluating image captions, particularly in the disaster context, we adopted the approach proposed by RSGPT [20]. This method assesses caption quality along two key dimensions: detailed description and localization accuracy. For each dimension, we implemented a four-tier rating system (A–D). Under this system, level A captions represent the highest quality. For detail, they provide comprehensive information

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TABLE IV

PERFORMANCE AND UPDATES FOR EACH SATELLITE ON EACH MISSION IN A THREE-SATELLITE SETUP

No.	Setting	Cycle-1(%)					Cycle-2(%)					Cycle-3(%)					AVG(CLS)	AVG(CAP)
Satellite A	without SGCL	29.7	25.7	74.5	74.2	46.0	29.8	25.1	73.3	77.1	44.0	30.2	25.5	72.5	77.1	48.0	65.2	27.6
	with SGCL	29.7	25.7	74.5	74.2	46.0	30.0	27.9	73.3	71.4	64.0†	30.6	27.9	72.6	77.1	76.0	69.9	28.6
Satellite B	without SGCL	37.5	71.4	51.2	72.2	30.1	40.0	68.5	53.6	71.9	29.4	35.0	71.4	53.6	71.8	29.8	58.2	29.7
	with SGCL	37.5	71.4	51.2†	72.2	30.1	57.5	68.5	63.4	71.9	29.4	72.5	71.4	63.4	71.9	29.8	64.4	29.7
Satellite C	without SGCL	72.6	29.9	80.4	24.9	32.3	73.3	29.6	80.4	23.0	37.1	73.6	29.8	82.9	24.8	25.7	62.8	27.0
	with SGCL	72.6	29.9	80.4	24.9	32.3†	73.1	29.8	80.4	25.0	45.7	73.4	30.1	82.9	26.7	48.5	65.3	27.7

TABLE V

PERFORMANCE AND UPDATES FOR EACH SATELLITE ON EACH MISSION IN A ONE-SATELLITE SETUP

No.	Setting	Cycle-1(%)					Cycle-2(%)					Cycle-3(%)					AVG(CLS)	AVG(CAP)
Satellite A	without SGCL	29.7	25.7	74.5	74.2	46.0	29.8	25.1	73.3	77.1	44.0	30.2	25.5	72.5	77.1	48.0	65.2	27.6
	with SGCL	29.7	25.7†	74.3	74.2	46.0†	30.2	27.8†	73.0	71.4	62.0†	30.0	28.2	72.2	77.1	76.0	69.6	28.6

TABLE VI

RELIABILITY VERIFICATION OF EXPERIMENTAL RESULTS

SGCL (seven satellites, annotation setting)	AVG(CLS)%	AVG(CAP)%
Learn then deploy	59.6 ± 0.15	28.6 ± 0.2
Annotation	65.8 ± 0.15	29.0 ± 0.05
Automated	63.1 ± 0.2	28.9 ± 0.15

TABLE VII

SGCL PERFORMANCE ACROSS FIVE SETTINGS AND WITH VARYING NUMBERS OF SATELLITES (EXPANDED BASELINES)

1. Setting (seven satellites)	**AVG-7 satellite A-G (CLS)**	**AVG-7 satellite A-G (CAP)**
Learn then deploy	59.8%	28.8%
FedGKD [13] (Online Distillation)	61.5%	28.9%
EWC-LoRA [37] (Continual Learning)	61.9%	28.8%
MPDA [12] (Domain Adaption)	62.1%	28.9%
FedRSCLIP [14] (Prompt Learning)	62.6%	29.0%
SGCL - Automated (Ours)	**63.3%**	**29.1%**
SGCL - Annotation (Ours)	**66.0%**	**29.1%**
2. Setting (three satellites)	**AVG-3 satellite A-C (CLS)**	**AVG-3 satellite A-C (CAP)**
Learn then deploy	61.8%	28.1%
FedGKD [13] (Online Distillation)	62.8%	28.3%
EWC-LoRA [37] (Continual Learning)	62.9%	28.2%
MPDA [12] (Domain Adaption)	63.1%	28.3%
FedRSCLIP [14] (Prompt Learning)	63.5%	28.3%
SGCL - Automated (Ours)	**64.1%**	**28.4%**
SGCL - Annotation (Ours)	**66.5%**	**28.6%**
3. Setting (seven satellites)	**AVG-3 satellite A-C (CLS)**	**AVG-3 satellite A-C (CAP)**
Learn then deploy	61.8%	28.1%
FedGKD [13] (Online Distillation)	63.2%	28.3%
EWC-LoRA [37] (Continual Learning)	63.4%	28.4%
MPDA [12] (Domain Adaption)	63.6%	28.4%
FedRSCLIP [14] (Prompt Learning)	63.9%	28.5%
SGCL - Automated (Ours)	**64.7%**	**28.4%**
SGCL - Annotation (Ours)	**67.3%**	**28.7%**
4. Setting (one satellite)	**satellite A (CLS)**	**satellite A (CAP)**
Learn then deploy	65.2%	27.6%
FedGKD [13] (Online Distillation)	65.8%	27.8%
EWC-LoRA [37] (Continual Learning)	65.9%	27.8%
MPDA [12] (Domain Adaption)	66.0%	27.8%
FedRSCLIP [14] (Prompt Learning)	66.3%	27.9%
SGCL - Automated (Ours)	**66.8%**	**28.0%**
SGCL - Annotation (Ours)	**69.6%**	**28.6%**
5. Setting (seven satellites)	**satellite A (CLS)**	**satellite A (CAP)**
Learn then deploy	65.2%	27.6%
FedGKD [13] (Online Distillation)	66.5%	27.8%
EWC-LoRA [37] (Continual Learning)	66.7%	27.8%
MPDA [12] (Domain Adaption)	66.9%	27.8%
FedRSCLIP [14] (Prompt Learning)	67.4%	27.9%
SGCL - Automated (Ours)	**68.2%**	**28.0%**
SGCL - Annotation (Ours)	**71.6%**	**28.7%**

TABLE VIII

EFFECT OF SGCL ON OOD PROBLEM, THE RESULTS ARE DERIVED FROM THE SGCL CONTAINING SEVEN SATELLITES IN THE ANNOTATION SETTING

OOD	Task	Before Update%)	After Update%)
Domain OOD	Agricultural Anomaly Detection	35.4	49.5
Domain OOD	Disaster Scenario Caption	24.8	26.5
Domain OOD	Disaster Scenario Relation Reasoning	38.5	53.3
Fine-grained OOD	Land Use Classification	54.4	61.5
Fine-grained OOD	Aircraft Type Classification	43.3	48.8
Fine-grained OOD	Ship Type Classification	36.7	46.3
Format OOD	Formal Shift	51.1	65.1
	Average	40.6	50.1

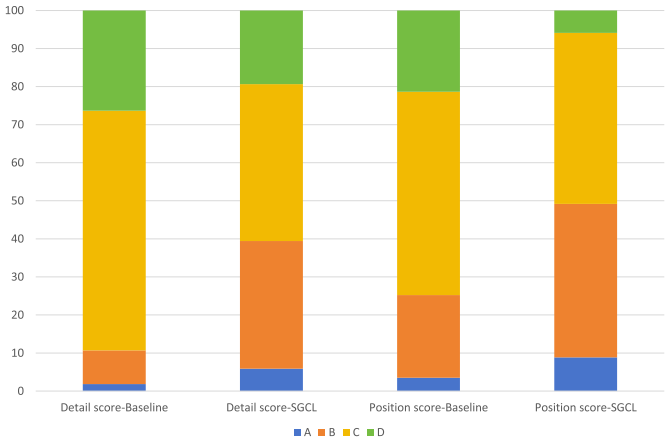


Fig. 3. Evaluation of the disaster caption task. The evaluation involves manual scoring based on detailed description and location description, is divided into four grades: A, B, C, or D, and is presented in the form of percentages.

capturing nearly all object types present; for localization, they deliver extensive and accurate positional descriptions for all objects. Level B captions also perform well, providing substantial detail covering most object types or detailed positional information with accuracy exceeding 50%. Level C represents adequacy, offering minimal detail describing only a few object types or either scant positional details or numerous positions but with accuracy below 50%. Finally, level D captions lack detail entirely or provide no positional descriptions at all. The scoring results across models (Fig. 3) demonstrate that SGCL significantly outperforms others, excelling particularly in capturing critical details and accurate spatial positioning.

To ensure the scientific rigor and objectivity of this manual grading system, three independent domain experts blindly rated the generated captions. To quantify their consensus, we computed Fleiss' Kappa for both evaluation dimensions. The statistical analysis yielded a Fleiss' Kappa of 0.78 for the

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detailed description dimension and 0.76 for the localization accuracy dimension. In statistical practice, values between 0.61 and 0.80 indicate substantial agreement among raters, thereby confirming that our manual evaluation protocol is consistent and reliable.

C. Comparative Evaluation With Other Methods

To justify the effectiveness of our proposed SGCL framework, we conducted a comparative evaluation that included a static "learn-before-deploy" baseline as well as four state-of-the-art lightweight collaborative and continual learning methods.

Because direct on-orbit training is physically prohibited by satellite power and thermal constraints, we implement all baseline methods under a strict server-simulated paradigm. Under this setup, the ground server executes the respective continual learning, online distillation, or prompt optimization algorithms using the downlinked OOD data and subsequently uplinks the lightweight updated weights to the satellite. This paradigm perfectly mirrors how SGCL updates the on-orbit model, ensuring a fair comparison.

Specifically, we incorporate EWC-LoRA [37] to represent lightweight continual learning, adapting the classic Elastic Weight Consolidation algorithm to the LoRA parameter space. We integrate FedGKD [13] to evaluate online global knowledge distillation, MPDA [12] to assess cloud-coordinated domain adaptation, and FedRSCLIP [14] for communication-efficient prompt learning tailored for remote sensing.

The comprehensive experimental results across five distinct evaluation settings are presented in Table VII. While all simulated baselines successfully outperform the static "learn then deploy" model, our proposed SGCL framework continues to maintain a substantial and consistent performance lead across all evaluation settings. We attribute this superiority to the fact that SGCL bypasses the fundamental limitations of these alternatives in dynamic aerospace environments.

When analyzing the performance gap against EWC-LoRA [37] and FedGKD [13], we note that both methods rely on conservative regularization. EWC applies rigid penalties to prevent forgetting, while FedGKD forces local models to mimic historical global ensembles. In highly dynamic orbital environments featuring unprecedented events, anchoring the model to historical normal distributions prevents rapid adaptation. SGCL overcomes this by leveraging the uncompressed foundation model to actively reason about the novel scenario, rather than dragging the on-orbit model back to historical states.

When comparing against MPDA [12], the analysis reveals its heavy reliance on retrieving similar data from a global cloud pool. In unpredictable orbital contexts like novel disaster mutations, the ground database often lacks sufficient similar source samples, causing domain adaptation to falter. SGCL bypasses this strict reliance on historical data retrieval through the foundation model's zero-shot generative reasoning capabilities via pseudo-labeling.

Finally, FedRSCLIP [14] utilizes prompt tuning, modifying only shallow input-level embeddings to save bandwidth. This approach lacks the internal representational capacity required to adapt to severe domain and fine-grained shifts. SGCL resolves this by utilizing parameter-efficient LoRA fine-tuning directly on the internal attention weights, thereby delivering significantly deeper representational plasticity and robustness.

D. Influence of Manual Annotation

The above results were measured under the data annotation setting, where anomaly detection triggers downlink transmission and human intervention provides annotation results. This manner achieved significant improvements, i.e., 6.2% classification accuracy gain with seven satellites and 4.7% classification accuracy gain with three satellites. As described in Section V-C, our framework also supports fully automated operation without manual intervention, relying exclusively on the ground foundation model. Even in automated mode, SGCL maintains solid performance leads. As shown in Table VII, under the seven-satellite setting, its classification accuracy still surpasses the baseline by 2.7%. Similarly, with only three satellites, automated operation also retains a 2.3% classification advantage. These consistent improvements demonstrate the versatility of our collaborative learning approach across operational modes.

E. Influence of Number of Satellites

Table VII evaluates how SGCL performance scales with varying satellite participation. The cross-satellite collaborative optimization effect is evidenced by comparing subexperiments. When Satellite A operates independently (Table VII.4), its annotation-mode classification accuracy is 69.6%. when collaborating with six peers (Table VII.5), its accuracy change to 71.6%.

As more satellites participate, they can collaborate through the ground foundation model, collecting more diverse data. This enables them to complement and optimize each other's task flows. Consequently, when other satellites in the system perform tasks with similar data distributions, they experience significant performance improvements, enhancing the accuracy and adaptability of the on-orbit models.

F. Effectiveness of OOD Detector

In the SGCL-bench, the OOD detector's identification of an OOD task is considered a successful prediction, and vice versa. As shown in Table III, all 32 OOD tasks were detected in the first two cycles. The OOD detection module incorporates a deduplication mechanism to suppress the transmission of repetitive abnormal data, thereby reducing communication overhead and model oscillation. Hence, during Cycle-3, it intentionally ignored 16 OOD tasks.

After removing visual reconstruction error, visual distribution shifts became undetectable, causing 22 OOD tasks to be detected in the first two cycles, while 10 were missed. The removal of the Mahalanobis distance component impairs the detection of text prompts shift, causing 20 OOD tasks to be detected in the first two cycles, while 12 were missed. Similarly, ablation of multimodal alignment prevents detection of cross-modal mismatches, resulting in 22 detected and ten undetected OOD tasks.

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TABLE IX
PERFORMANCE COMPARISON OF SGCL FRAMEWORK COMPONENTS

A. SGCL (seven satellites, annotation setting)	AVG (CLS)	AVG (CAP)
On-orbit model	66.0%	29.1%
Foundation model	75.8%	35.2%
B. Delete foundation model update	**AVG (CLS)**	**AVG (CAP)**
On-orbit model	64.4%	28.8%
Foundation model	73.3%	33.8%
C. Delete "adaptive" in distillation	**AVG (CLS)**	**AVG (CAP)**
On-orbit model	64.8%	28.9%
Foundation model	74.0%	34.1%

TABLE X
ROBUSTNESS EVALUATION UNDER NOISE AND ADVERSARIAL SHIFTS

Test Condition	Perturbation Type	Metric	Trigger Rate
Clean Baseline	No perturbation (In-Distribution)	False Positive Rate	3.2%
Real Satellite Noise	Gaussian Sensor Noise ($\sigma = 25$)	False Positive Rate	4.8%
Real Satellite Noise	Synthetic Cloud Disturbance	False Positive Rate	5.5%
Adversarial Shift	PGD Visual Attack ($\epsilon = 8/255$)	**True Positive Rate**	89.4%

To further evaluate robustness under realistic conditions, we conducted a controlled experiment against real satellite noise and adversarial shifts. As shown in Table X, under severe Gaussian sensor noise (σ = 25) and synthetic cloud disturbances, the false positive rate (FPR) remains below 6%, confirming the system's stability against natural physical perturbations. Conversely, under a pixel-level projected gradient descent adversarial attack (ϵ = 8/255), the true positive rate (TPR) reaches 89.4%, demonstrating high sensitivity to malicious out-of-manifold feature perturbations. These results validate that our OOD Detector effectively distinguishes between benign noise and significant distribution shifts.

G. Effectiveness of Update Trigger

As shown in Table III, under the setup with seven satellites, our SGCL performed 28 on-orbit model updates and 4 ground model updates in the first two cycles, accounting for 40% and 5.7% of all tasks, respectively. As shown in Table IV, with three satellites, our SGCL conducted ten on-orbit model updates and three ground model updates in the first two cycles, representing 33.3% and 10% of tasks, respectively.

We determine update triggering by evaluating distributional discrepancies between on-orbit model predictions and foundation model outputs, using the on-orbit model's task performance as the criterion to validate update trigger accuracy—where updates are only activated upon on-orbit performance degradation. As demonstrated in Table III's seven-satellite scenario, all updates occurred exclusively during on-orbit performance decline. For four specific satellite task cases, while the OOD module triggered data downlink due to anomalies, the update trigger rejected model updates after assessing prediction consistency (discrepancies between ground and on-orbit models remained below threshold). Subsequent performance of the satellite models in these four cases, as evidenced in Table III, confirms the validity of this decision.

H. Performance on Multiple OOD Tasks

As shown in Table VIII, we quantify SGCL's efficacy against four categories of OOD challenges. Our framework consistently elevates task performance post-update across all seven scenarios: domain OOD tasks (e.g., agricultural anomaly detection improved from 35.4% to 49.5%) and fine-grained OOD recognition (e.g., ship type classification surged from 36.7% to 46.3%).

Specifically, to verify the effects of prompt variations on VLM outputs, we evaluated the framework on the format OOD task. Unadapted on-orbit VLMs are highly sensitive to user prompt perturbations (e.g., complex syntax or ambiguous specifications), resulting in a low initial accuracy of 51.1%. By detecting these textual prompt shifts through the OOD module and triggering collaborative LoRA updates, our SGCL framework successfully mitigated the negative effects of prompt variations, elevating the formal shift accuracy to 65.1%.

Overall, the average performance gain of 9.5% across all OOD scenarios demonstrates SGCL's decisive advantage over static "learn then deploy" paradigms during highly dynamic on-orbit inference.

I. Influence of Foundation Model Update

We conducted ablation experiments to analyze the impact of unidirectional knowledge transfer (only from the ground model to the satellite model) versus bidirectional asynchronous updates and to measure the performance improvement the foundation model gains from the SGCL framework. As shown by comparing parts A and B in Table IX, removing updates to the ground foundation model leads to a significant performance degradation in both models: The on-orbit model's performance (both CLS and CAP metrics) decreases, demonstrating its reliance on knowledge transferred from the continuously improving foundation model; the foundation model's performance itself declines substantially, such as avg. CLS drops from 75.8% to 73.3%. This quantitatively confirms that SGCL's mechanism for incorporating knowledge from the on-orbit models provides a clear performance boost to the ground foundation model.

J. Influence of Adaptive Distillation

Comparing parts A and C in Table IX, the results demonstrate a concrete benefit from the adaptive distillation strategy. When we remove the ability to distinguish and then receive knowledge from different on-orbit models, both the foundation model and the on-orbit model show decreased performance. This confirms that adaptively integrating diverse knowledge sources from the various on-orbit satellites, rather than treating them uniformly, is advantageous for the system overall.

K. Qualitative Comparison

We present qualitative results in Fig. 4. The categories of airplanes and ships fall under fine-grained OOD, the categories of VQA-formal shift fall under format OOD, and the categories of Relation Reasoning-Disaster fall under domain OOD. Additionally, we found that updates to the on-orbit

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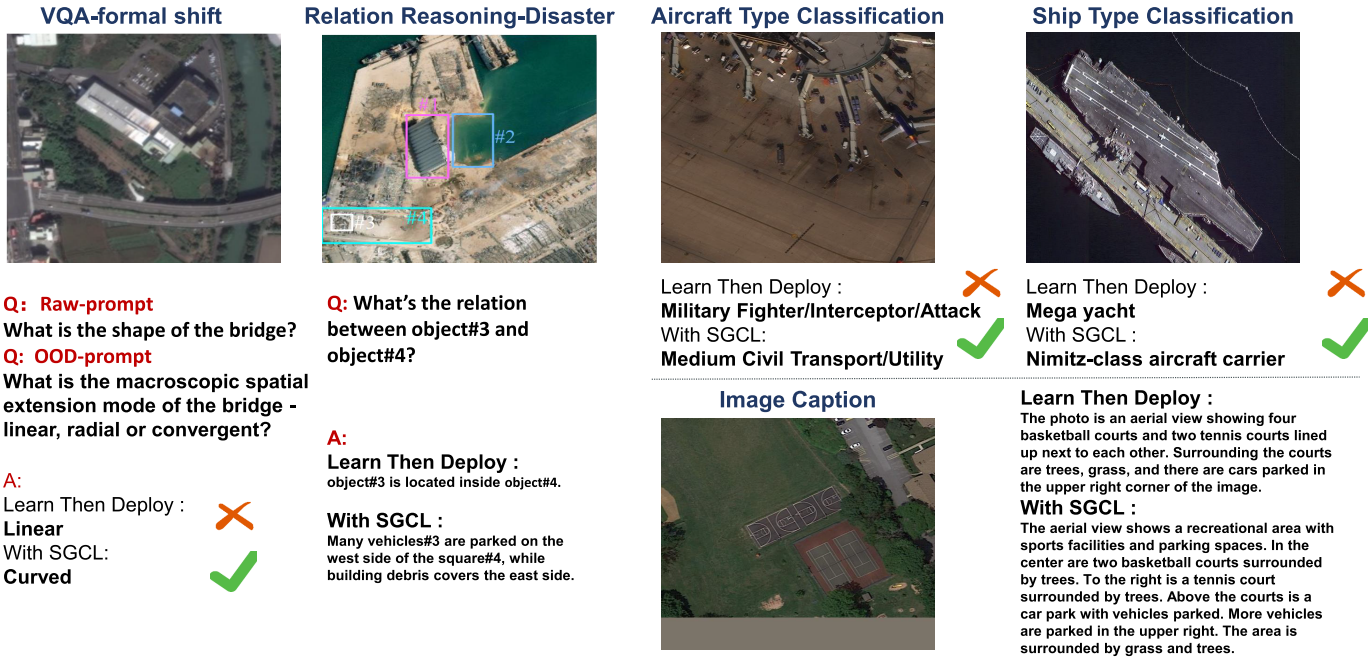


Fig. 4. Qualitative comparison.

model triggered by other tasks can lead to more detailed image captions.

L. Validation and Limitation

For the on-orbit model (Qwen2-VL-2B), a single update using LoRA technology requires no more than 7 M parameters. The resulting data transfer size is approximately 13.4 MB (calculated at FP16 precision). Including overhead for checksums, metadata, and other protocol elements, the actual satellite-to-ground transmission size remains under 15 MB per update. A typical satellite pass lasts 5–10 min (300–600 s). Even with a low-rate downlink (5 Mb/s), the transmission takes only 25 s, accounting for just 4.2%–8.3% of the available pass time. We have successfully validated this on-orbit model update capability via uplink on the Oriental Smart Eye 01 Satellite.

Our proposed SGCL framework explicitly considers the limitations inherent in existing space systems, such as low-bandwidth communication and constrained on-orbit computing resources. We have successfully validated the framework and satellite–ground interaction using data from China's Oriental Smart Eye 01 Satellite. However, comprehensive multisatellite constellation testing remains limited to benchmark evaluation (SGCL-Bench) due to practical resource constraints. Future work will target collaboration with organizations operating multisatellite constellations to enable on-orbit validation of the full SGCL framework.

Furthermore, practical satellite–ground systems frequently operate under severe annotation budgets and communication limits. Recent studies have demonstrated significant potential in adapting powerful foundation models under scarce data conditions, such as combining the segment anything model (SAM) with limited data for remote sensing change detection [38] and exploring efficient open-vocabulary segmentation [39]. Integrating these data-efficient adaptation strategies into our SGCL framework represents a highly promising future extension. This integration could further minimize manual annotation reliance and significantly reduce the computational costs of edge-side perception tasks.

VI. Conclusion

We propose the SGCL framework, a novel solution addressing on-orbit OOD challenges. SGCL leverages bidirectional asynchronous updates between compressed satellite models and a ground-based foundation model. To facilitate comprehensive evaluation under realistic conditions, we also propose the SGCL Benchmark, simulating multisatellite scenarios with cyclical workflows involving seven satellites across seven tasks covering domain, fine-grained, and format OOD. Extensive experiments demonstrate SGCL's significant adaptability enhancement, achieving an average performance gain of 9.5% over baselines across all seven OOD scenarios. This work highlights the substantial potential of collaborative learning for advancing remote sensing applications.

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